

DSA + System Design + Behavioural

Two candidates get the same system-design question. Candidate A dives into model architecture in minute 2. Candidate B spends 20 minutes on clarification + metrics + data. Candidate B gets the offer. The best candidates think BEFORE they architect.

When did you last spend 20 minutes clarifying before building?

WHAT'S INSIDE

- 01** 4 rounds
- 02** DSA — 150 mediums
- 03** System design — 7 steps
- 04** STAR for behavioural
- 05** Debugging scenario — framework

WHY THIS LESSON MATTERS

Same candidate. Two outcomes. Based on clarification alone.

NUMBER 1

20 min

From real-world deployments.

NUMBER 2

20 min

From real-world deployments.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

4 interview rounds.

2

Medium DSA problem in 25 min.

3

ML system design in 45 min.

4

Behavioural using STAR.

5

Debug an ML scenario.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

4 rounds

Key concept of this lesson.

02

DSA — 150 mediums

Patterns that matter

03

System design — 7 steps

Never skip the first two

04

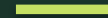
STAR for behavioural

6 stories, reusable

01

PART 1 OF 4

4 rounds



Key concept of this lesson.

CONCEPT 1 OF 4

4 rounds

A core idea you'll use throughout this lesson.

DEFINITION

4 rounds

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

DSA — 150 mediums

Arrays + hashmap (30% of asks).

Two-pointer, sliding window.

Binary search (sorted + on answers).

Graph BFS/DFS.

DP with memoisation.

PREP

150 mediums

Across the 6 patterns.

KEY POINTS

- Arrays + hashmap (30% of asks)
- Two-pointer, sliding window
- Binary search (sorted + on answers)
- Graph BFS/DFS

CONCEPT 3 OF 4

System design — 7 steps

1. Clarify (5 min) — scale, latency, accuracy, budget.
2. Metrics (5 min) — business + model.
3. Data (10 min) — source, freshness, leakage.
4. Model (10 min) — baseline first.
5. Serving (5 min) — latency budget.
6. Monitoring (5 min) — drift, SLOs.
7. Tradeoffs (5 min) — summary.

ORDER

1 → 7

Skipping 1-2 = failed interviews.

DEFINITION

System design — 7 steps

Never skip the first two

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

STAR for behavioural

Situation — 1-2 sentence context.

Task — what you were asked.

Action — what YOU did.

Result — quantified + learning.

PREP

6 stories

Adapt to any question.

KEY POINTS

- Situation — 1-2 sentence context
- Task — what you were asked
- Action — what YOU did
- Result — quantified + learning

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

30-min system design mock

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

- 1
- 2 30-min system design mock
- 3
- 4 Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Which round is your weakest?

Q2

Which of your 6 STAR stories is rehearsed?

Q3

Where do you jump to solution too fast?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

4 rounds:



A



DSA, ML fundamentals, system design, behavioural



Only DSA



Only behavioural

4 rounds:



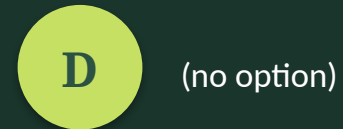
CORRECT ANSWER

DSA, ML fundamentals, system design, behavioural

WHY

Typical AI-engineer interview loop

Minimum Leetcode prep:



Minimum Leetcode prep:



CORRECT ANSWER

E

WHY

~150 across 6 patterns

Biggest system-design mistake:

A

Too slow

B

Jumping to model in minute 2

C

Not smiling

D

Fast typing

Biggest system-design mistake:



CORRECT ANSWER

Jumping to model in minute 2

WHY

Skips clarification

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

A team applies the lesson's main idea to a real workflow — and the results show up within a quarter.

THE TAKEAWAY

"Quantify before you commit."

WHAT TO NOTICE

- 1 4 rounds
- 2 DSA — 150 mediums
- 3 System design — 7 steps
- 4 STAR for behavioural

Prep 6 STAR stories

From W1-W7 projects.

DELIVERABLES

- 6 stories.
- Pair mock 30 min.
- Feedback notes.
- Iterate.

WHAT 'GOOD' LOOKS LIKE

Deliverable: 6 stories + mock feedback.

ONE THING TO REMEMBER

“

"Quantify before you commit."

— AI Learning Hub

END OF LESSON 8

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 7 — Salary Negotiation.